

# WaziGov

An AI-Powered Transparency System for Public finance  
Monitoring



Computer Vision



NLP & Machine Learning



Geospatial Intelligence

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# 1. INTRODUCTION AND BACKGROUND

## Context & Motivation

Kenyan citizens have a constitutional right to know how public money is spent and raise questions where necessary.

### The Data Trap

While the files are digital, the **structure is complex**. A citizen would find it difficult to understand the complex documents.

## Core Vision

A unified pipeline that:

**Recovers Structure** (CV)

**Extracts Meaning** (NLP)

**Maps Context** (Geo)

## 2. PROBLEM STATEMENT

### Broader Computing Problem

#### The "Cross-Modal Integration" Challenge

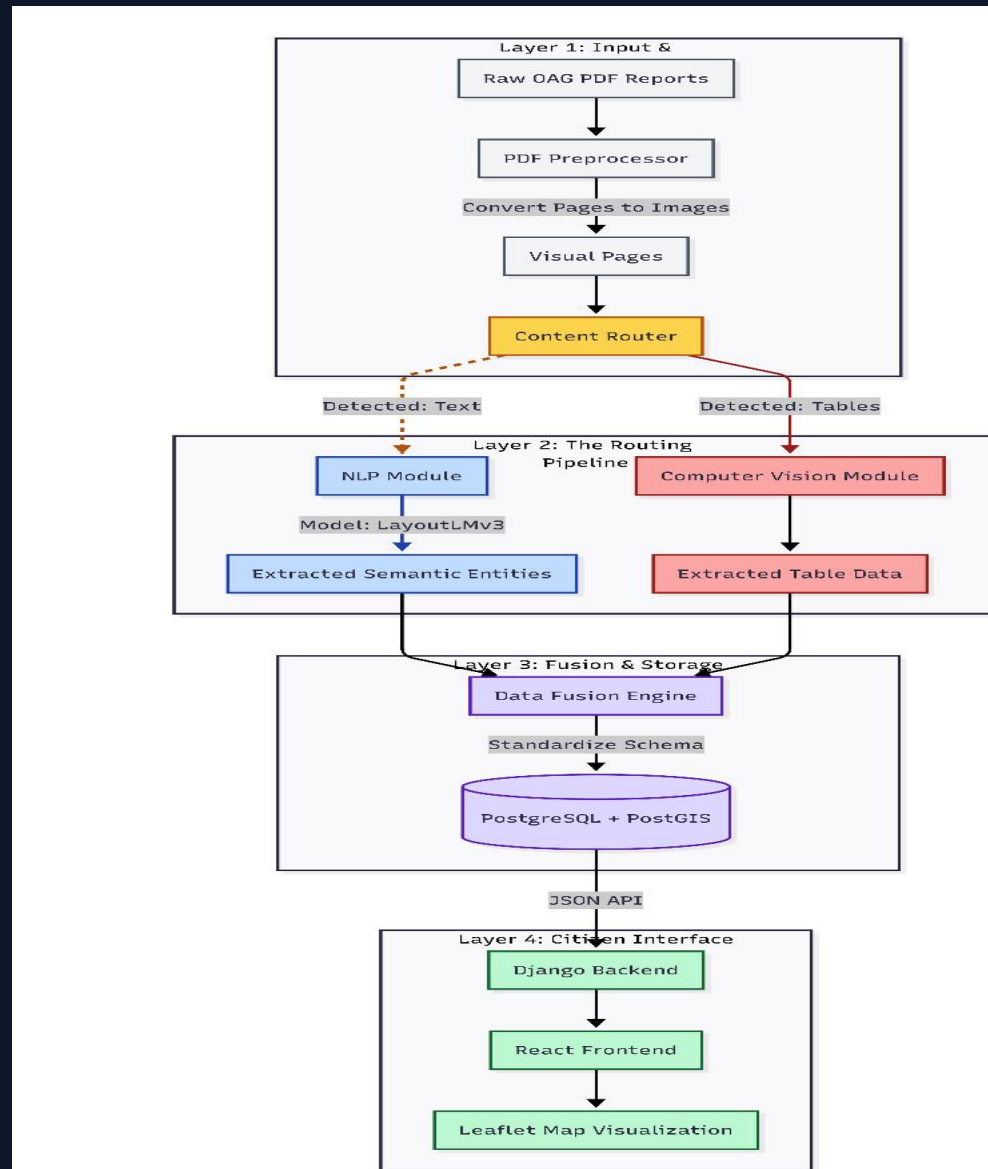
The main issue is that a document's structure, its language, and its geographic context do not talk to each other. Our technical challenge is to bridge these gaps automatically, turning scattered data into a single, searchable reality

### Real World Use Case

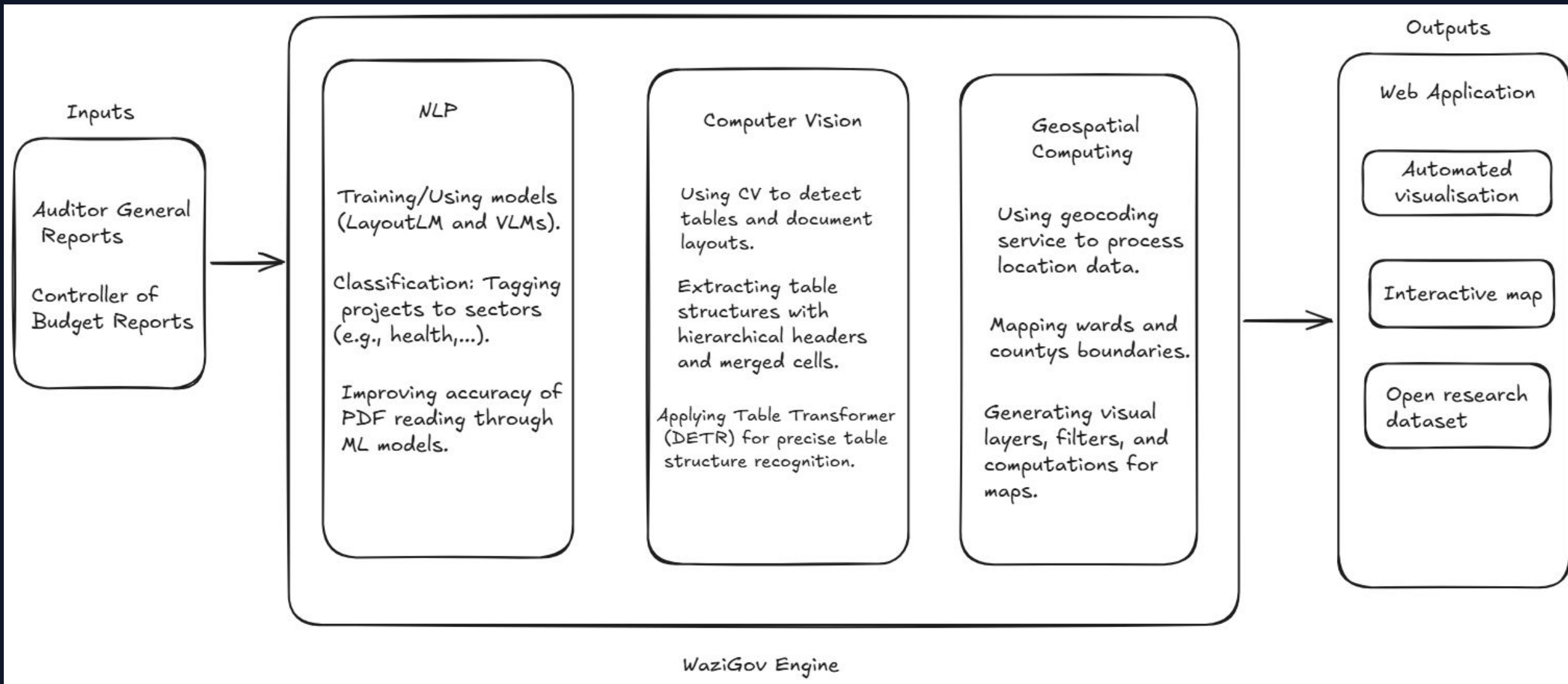
#### The "Accessibility Gap"

This system bridges the "**Accessibility Gap**" by transforming static, technical reports into interactive tools that empower citizens to easily verify public projects and usage of their taxes.

### 3. PROPOSED SOLUTION: SYSTEM ARCHITECTURE



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# 3.1 TECHNICAL ROLES

| Role & Domain                                   | Focus                               | Core Responsibilities                                                                                                                                                                                                                                      | Technology Stack                                                 |
|-------------------------------------------------|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| Computer Vision Financial Document Intelligence | <b>STRUCTURE</b><br>(Input & Logic) | <ul style="list-style-type: none"><li>· <b>Hierarchy:</b> Handling 3+ level headers &amp; merged cells.</li><li>· <b>Continuity:</b> Stitching tables spanning multiple pages.</li><li>· <b>Validation:</b> IPSAS-based arithmetic logic checks.</li></ul> | Table Transformer (DETR), CascadeTabNet, LGPMA, Custom Validator |
| Machine Learning Natural Language Processing    | <b>CONTENT</b><br>(The Processing)  | <ul style="list-style-type: none"><li>· <b>Entity Extraction:</b> Parsing text from grid coordinates.</li><li>· <b>Cleaning:</b> Normalizing currency &amp; dates.</li><li>· <b>Classification:</b> Tagging projects (e.g., "Health").</li></ul>           | BERT Transformers, Pandas, Regex SQL, LayoutLM                   |
| Geospatial Location Intelligence                | <b>CONTEXT</b><br>(The Output)      | <ul style="list-style-type: none"><li>· <b>Geocoding:</b> Linking location names to GPS.</li><li>· <b>Mapping:</b> Assigning projects to Administration Wards.</li><li>· <b>Visualization:</b> Interactive Dashboard.</li></ul>                            | PostGIS Leaflet / Mapbox QGIS React                              |

## 4. MAIN OBJECTIVE

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To develop an AI-driven framework that extracts financial entities from government reports and projects them into interactive administrative maps for public monitoring and easy interaction.



## 4.1 SPECIFIC OBJECTIVES

01

### Visual Extraction Pipeline

To adapt existing Visual Language Models and Document Layout Analysis techniques for processing structurally complex government PDF reports with hierarchical tables, merged cells, and cross-page continuation.

02

### Intelligent Data Processing

To design NLP algorithms that clean, classify, and validate extracted entities, transforming raw strings into structured financial data.

03

### Geospatial Dashboard

To develop an interactive platform that links fiscal data to administrative boundaries, visualizing the "Accessibility Gap" for citizens.

| 5. STATE OF THE ART: RESEARCH LANDSCAPE(CV)                                                                                                         |                                                               |                                                                                     |                                                                              |                                                          |
|-----------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------|----------------------------------------------------------|
| Method/Approach                                                                                                                                     | Key Innovation                                                | Strengths                                                                           | Limitations                                                                  | Relevance to WaziGov                                     |
| <i>DeepDeSRT: Deep Learning for Detection and Structure Recognition of Tables in Document Images</i> (Schreiber et al., 2017)                       | First end-to-end deep learning with VGG-16 + RPN              | 91.4% F1 on ICDAR 2013                                                              | Separate models for detection/structure; slow (5-10s/page)                   | Demonstrates feasibility of deep learning for tables     |
| <i>TableNet: Deep Learning Model for End-to-End Table Detection and Tabular Data Extraction from Scanned Document Images</i> (Paliwal et al., 2020) | Unified encoder-decoder with VGG-19, semantic segmentation    | End-to-end architecturer                                                            | Struggles with precise cell boundaries; merged cells not explicitly modelled | Highlights need for explicit merged cell handling        |
| DETR<br>End-to-end object detection with transformers(Carion et al., 2020)                                                                          | Detection Transformer - direct set prediction, no anchors/NMS | Eliminates hand-crafted components; self-attention captures long-range dependencies | Computationally intensive                                                    | Foundational architecture for modern table understanding |
| Table Transformer<br>PubTables-1M: Towards Comprehensive Table Extraction from Unstructured Documents(Smock et al., 2022)                           | DETR adaptation for tables; two-stage detection + structure   | 96.6% precision, 95.4% recall; 94.2% GriTS; pre-trained on 947K tables              | Trained on scientific papers (different domain)                              | BASE MODEL FOR THIS PROJECT                              |
| <i>LGPMMA: Complicated Table Structure Recognition with Local and Global Pvramid Mask Alignment</i> (Qiao et al., 2022)                             | Local/Global Pyramid Mask Alignment for merged cells          | Handles complicated table structures                                                | Complex architecture                                                         | Informs merged cell detection approach                   |

# 5. STATE OF THE ART: RESEARCH LANDSCAPE (NLP)

| Method/Approach                                                                                                                | Key Innovation                                                            | Strengths                                                                                        | Limitations                                                                                             | Relevance to WaziGov                                                                              |
|--------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| LayoutLMv3 (Yupeng Huang et al., 2022)                                                                                         | Unified pre-training of text, layout (coordinates), and image modalities. | Captures spatial relationships; understands that position determines meaning in structured PDFs. | High computational cost and VRAM usage.                                                                 | Essential for linking fiscal findings to their visual context and section headers in OAG reports. |
| A BERT-based Hierarchical Classification Model (Kun Liu et al., 2025)                                                          | Hierarchical fine-tuning strategy for structured classification tasks.    | Leverages taxonomy-aware fine-tuning; strong performance on hierarchical text classification.    | Not originally designed for long-document financial reasoning; no explicit sliding-window architecture. | Conceptually informs structured classification of multi-level administrative audit data.          |
| Enhancing Hierarchical Classification in Tree-Based Models Using Level-Wise Entropy Adjustment (Olga Narushynska et al., 2025) | Penalized Information Gain (PIG) for hierarchy-aware decision trees.      | Maintains taxonomic integrity; balances entropy reduction with hierarchical coherence.           | Requires custom loss calibration and hierarchy-aware tuning.                                            | Ensures audit findings are correctly mapped to Kenya's 47 counties without ancestry violations.   |
| TabFact: A Large-scale Dataset for Table-based Fact Verification (Wenhu Chen et al., 2020)                                     | Symbolic-neural integration for table-based fact verification.            | Reduces hallucination by reconciling claims against structured numeric tables.                   | Complex alignment between natural language statements and executable logic.                             | Foundation for a structured “JSON Contract” layer to verify audit prose against tabular annexes.  |
| PLTA-FinBERT: Pseudo-Label Generation-Based Test-Time Adaptation for Financial Sentiment Analysis (Hai Yang et al., 2026)      | Test-time adaptation via pseudo-label generation.                         | Maintains performance under semantic drift in financial language.                                | Risk of instability if pseudo-labels are noisy.                                                         | Supports longitudinal analysis (2018–2024) amid evolving public financial reporting language.     |

## 5. STATE OF THE ART: RESEARCH LANDSCAPE

| Title / Focus                                                                                                | Key Features & Interaction                                                                                                                      | Main Benefits / Findings                                                                                                                               | Limitations / Requirements                                                                          | Relevance to WaziGov / Project Context                                                                                                                                     |
|--------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Interactive Web-Based Visualization of Multidimensional Physical and Astronomical Data [Diblen et al., 2021] | SPOT framework for 1-3D visualization & histograms of high-dim data; client/server workflows; supports CSV/JSON; dynamic linking between plots. | Enables quick reanalysis & comparison of phenomenological data; facilitates open science & data reuse; allows non-experts to find new correlations.    | Requires technical expertise for existing tools (ROOT, Matplotlib); generic tools lack specificity. | Demonstrates a framework for interactive exploration of complex, multi-parameter datasets, relevant for visualizing multifaceted ward-level financial or demographic data. |
| Interactive Visualization and Representation Analysis Applied to Glacier Segmentation [Zheng et al., 2022]   | Shiny app with linked map & accuracy curve; representation analysis via U-Net activation visualization; focusing & linking techniques.          | Guides model interpretation; reveals model failure modes (e.g., incomplete labels); provides sanity checks for data preprocessing.                     | Model performance limited by data quality (e.g., incomplete labels).                                | Supports model interpretability and error analysis for any ML-based predictions within WaziGov (e.g., predicting service delivery needs), ensuring transparency.           |
| Geographic Vector Big Data Based on Viewport Generalization Model [Chen et al., 2022]                        | Viewport generalization model (VGVM); Viewport Pixel Quadtree (VPQ) index; pixel-as-unit computation; parallel algorithm (VPQP).                | Converts data-scale complexity to pixel-scale complexity; achieves >10x performance improvement; enables real-time preview for billion-scale datasets. | Performance may degrade when viewport pixel size is larger than data feature size.                  | Provides a core technology for fast, interactive rendering of large vector datasets (e.g., ward boundaries, infrastructure lines) in a web-based WaziGov dashboard.        |
| A Display-driven Spatial Analysis Framework for Large-Scale Vector Data (HiVision) [Ma et al., 2021]         | Display-driven visualization; takes pixels as computational units; queries vector elements affecting each pixel.                                | Faster tile-generation than HadoopViz & GeoSparkViz; supports real-time visualization of big geographic vector data (BGVD).                            | Implementation details are complex, requiring specific spatial indexes.                             | Offers a high-performance alternative for rasterizing large geospatial datasets, which could be used to create fast, interactive base maps for WaziGov.                    |

# 5. STATE OF THE ART: RESEARCH LANDSCAPE

| Method/Approach                                                         | Key Innovation                            | Strengths                        | Limitations               | Relevance to WaziGov                            |
|-------------------------------------------------------------------------|-------------------------------------------|----------------------------------|---------------------------|-------------------------------------------------|
| Viewport-based geospatial rendering [Chen, L., Liu, Z., & Ma, M., 2022] | Efficient rendering of large map datasets | Improves performance of web maps | Requires spatial indexing | Relevant for ward-level financial map rendering |

# 6. METHODOLOGY

An end-to-end framework integrating deep learning, NLP, and geospatial systems.



## PHASE 1

### Data Collection & Annotation

Collect OAG and Controller of Budget reports. Manually annotate bounding boxes, merged cells, and headers to create a robust fine-tuning dataset for government documents.



## PHASE 2

### CV Pipeline Development

Fine-tune Table Transformer for structure recognition. Integrate CascadeTabNet for borderless table detection and LGPMA-informed post-processing for complex merged cells.



## PHASE 3

### NLP Processing

Deploy LayoutLMv3 for header/data cell classification. Apply BERT-based entity extraction, cleaning, and sector classification to structural outputs.



## PHASE 4

### Geospatial Integration

Geocode extracted location entities and map them to ward boundaries via PostGIS. Build an interactive citizen dashboard using Leaflet.



## PHASE 5

### Evaluation & Testing

Measure CV accuracy using GrITS and F1 scores. Validate NLP classification accuracy and conduct user testing on the citizen dashboard.

## 7. EXPECTED OUTCOMES



### Open Research Dataset

A clean, exportable dataset (CSV) made available for use by other researchers and developers.



### Open Source Toolkit

A reusable Python library specifically tuned for “Document Layout Analysis” of Kenyan government financial documents.



### Citizen Dashboard

An interactive web platform enabling citizens to select their Ward and verify funded projects, closing the accountability loop.

Impact: Transforming "Read-Only" government data into "Read-Write" accountability.

# 8. Project Execution Timeline

A strategic 19-week roadmap from research to deployment.





## 9. Responsible Computing Aspects

| Aspect               | Strategy / Approach       | Outcome & Impact                                                                                     |
|----------------------|---------------------------|------------------------------------------------------------------------------------------------------|
| <b>Data Privacy</b>  | Open Architecture Design  | Eliminates the need for user PII (Personally Identifiable Information), ensuring privacy by default. |
| <b>Inclusivity</b>   | Screen Reader Integration | Ensures every citizen, including the blind and visually impaired, is informed and included.          |
| <b>Fairness</b>      | No Account Creation       | Prevents profiling and ensures information is fed in equal measures to all users without bias.       |
| <b>Social Impact</b> | Transparent Reporting     | Brings accountability to Public Finance and Management by democratizing data access.                 |
| <b>Safety</b>        | Non-Tracking Architecture | Reduces suspicion and avoids the perception of tracking when interacting with the system.            |

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## Geovisualization Techniques of Spatial Environmental Data Using Different visualization Tools

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**THANK YOU**